



**American
Red Cross**

Carnegie Mellon University
HeinzCollege
INFORMATION SYSTEMS • PUBLIC POLICY • MANAGEMENT

American Red Cross: Geospatial Modeling of Real Time Earthquake Impacts to Housing

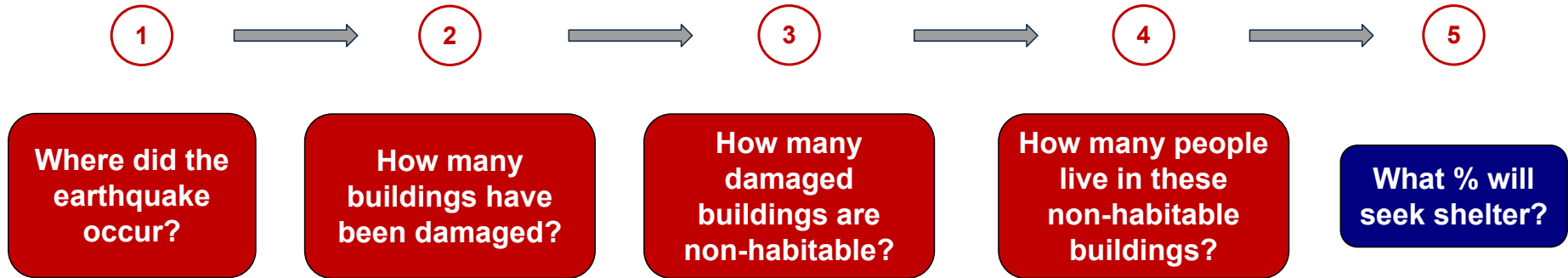
Brief Overview

Challenge: Mass Care Emergency Planning

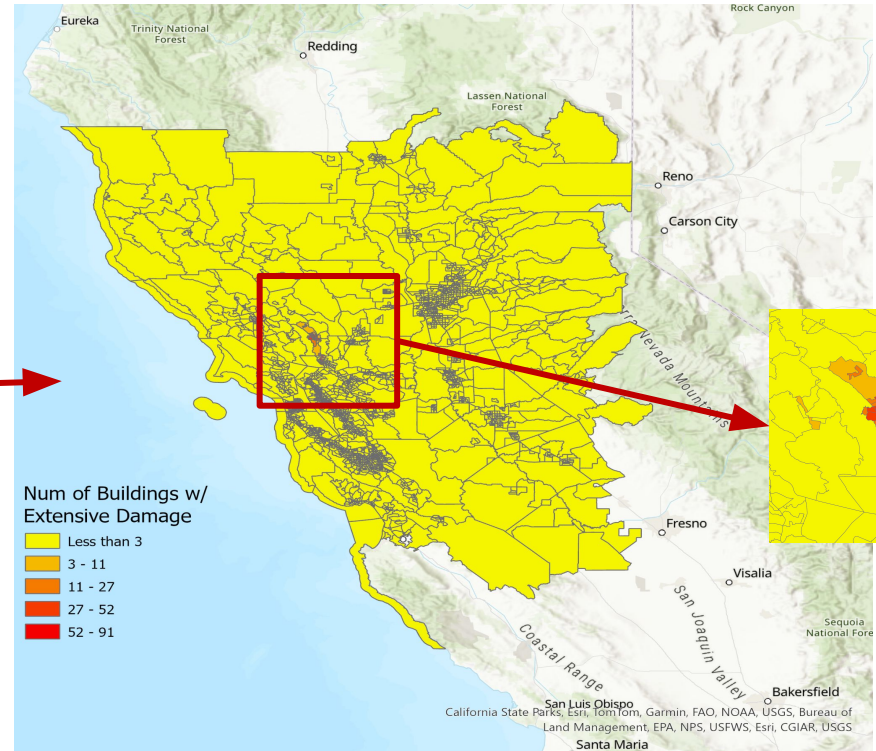
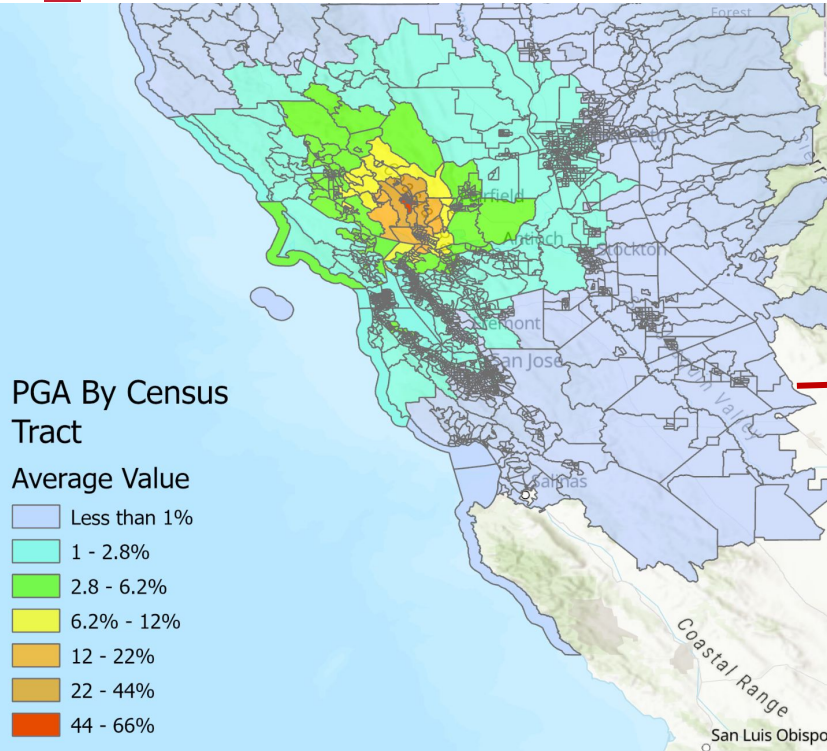
- **Earthquake occurs**
- The American Red Cross needs to start mobilizing resources asap (under an hour).
- To do so, it needs to know how many people will need shelter.
- Current tools provide the data 3 days after the earthquake.



Methodology Workflow



Model 1: Estimating Structural Damage



Model 2: Estimating Shelter Seeking Behavior

Loss of Utility Services

- Power
- Water
- Sewage

Social Vulnerability

Overall Vulnerability

Socioeconomic Status

Below 150% Poverty

Unemployed

Housing Cost Burden

No High School Diploma

No Health Insurance

Aged 65 & Older

Aged 17 & Younger

Civilian with a Disability

Single-Parent Households

English Language Proficiency

Household Characteristics

Racial & Ethnic Minority Status

Hispanic or Latino (of any race)
Black or African American, Not Hispanic or Latino
Asian, Not Hispanic or Latino
American Indian or Alaska Native, Not Hispanic or Latino
Native Hawaiian or Pacific Islander, Not Hispanic or Latino
Two or More Races, Not Hispanic or Latino
Other Races, Not Hispanic or Latino

Housing Type & Transportation

Multi-Unit Structures

Mobile Homes

Crowding

No Vehicle

Group Quarters

Results

Validation

2014 Napa Valley (M6.0)

Red Cross Reported Peak Shelter Usage	52
Our Prediction	48 - 204
Pager Impacted Population (Available Shortly After Earthquake)	77,000 (For Napa)

2022 Humboldt (M6.4)

Red Cross Reported Peak Shelter Usage	13
Our Prediction	22 - 78
Pager Impacted Population (Available Shortly After Earthquake)	3,000

Appendix

General Building Stock

What is it?

- Census Tract-level breakdown of structure types (material and building code)

How We Use It?

- Capturing **building structure type composition** by census tract
- Feed data into damage estimation models.

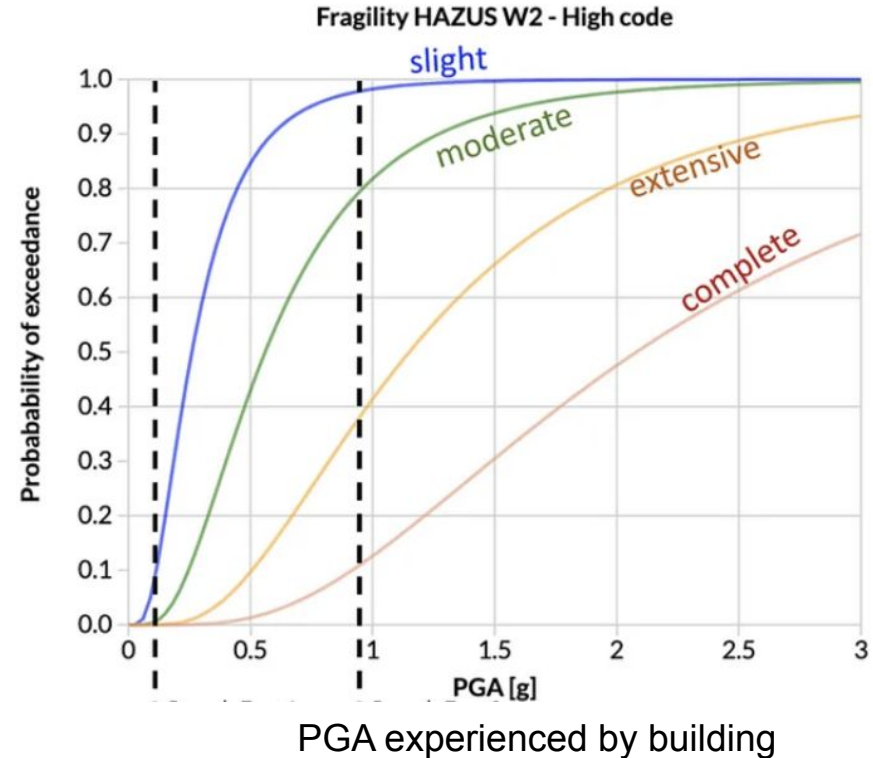
W2:

- Commercial, industrial, or multi-family residential buildings exceeding 5,000 sq. ft.
- **Wood Structural and Lateral Load Resistance Design**

Tract	W1	W2	S1L	S1M	S1H
6001400100	93.12%	1.12%	0.49%	0.00%	0.00%
6001400200	88.04%	1.87%	1.00%	0.00%	0.00%
6001400300	86.18%	2.08%	0.95%	0.00%	0.00%
6001400400	88.71%	1.58%	0.58%	0.00%	0.00%
6001400500	87.30%	1.48%	0.56%	0.00%	0.00%
6001400600	88.55%	1.07%	0.54%	0.00%	0.00%
6001400700	86.18%	1.59%	0.80%	0.00%	0.00%

Hazus Fragility Curves

- For each **building type** and **level of damage**, HAZUS provides a function expressing the cumulative probability of exceeding a damage state given a level of Peak Ground Acceleration (PGA) → **Cumulative Damage Probability (CDP)**



Hazus Fragility Curves

$$CDP \left(\frac{1}{\beta} \ln \left(\frac{PGA}{\text{median PGA}} \right) \right)$$

Assumptions we make:

All buildings in a county are assigned the **min PGA** observed in that county

Cumulative Damage Probability (CDP) parameterized by:

- **PGA:** PGA experience by building
- **Beta:** represents the variability in how buildings respond to shaking → $1/\text{Beta}$ is a scaling factor
- **Median PGA:** the shaking level at which a building has a 50% chance of reaching the damage threshold